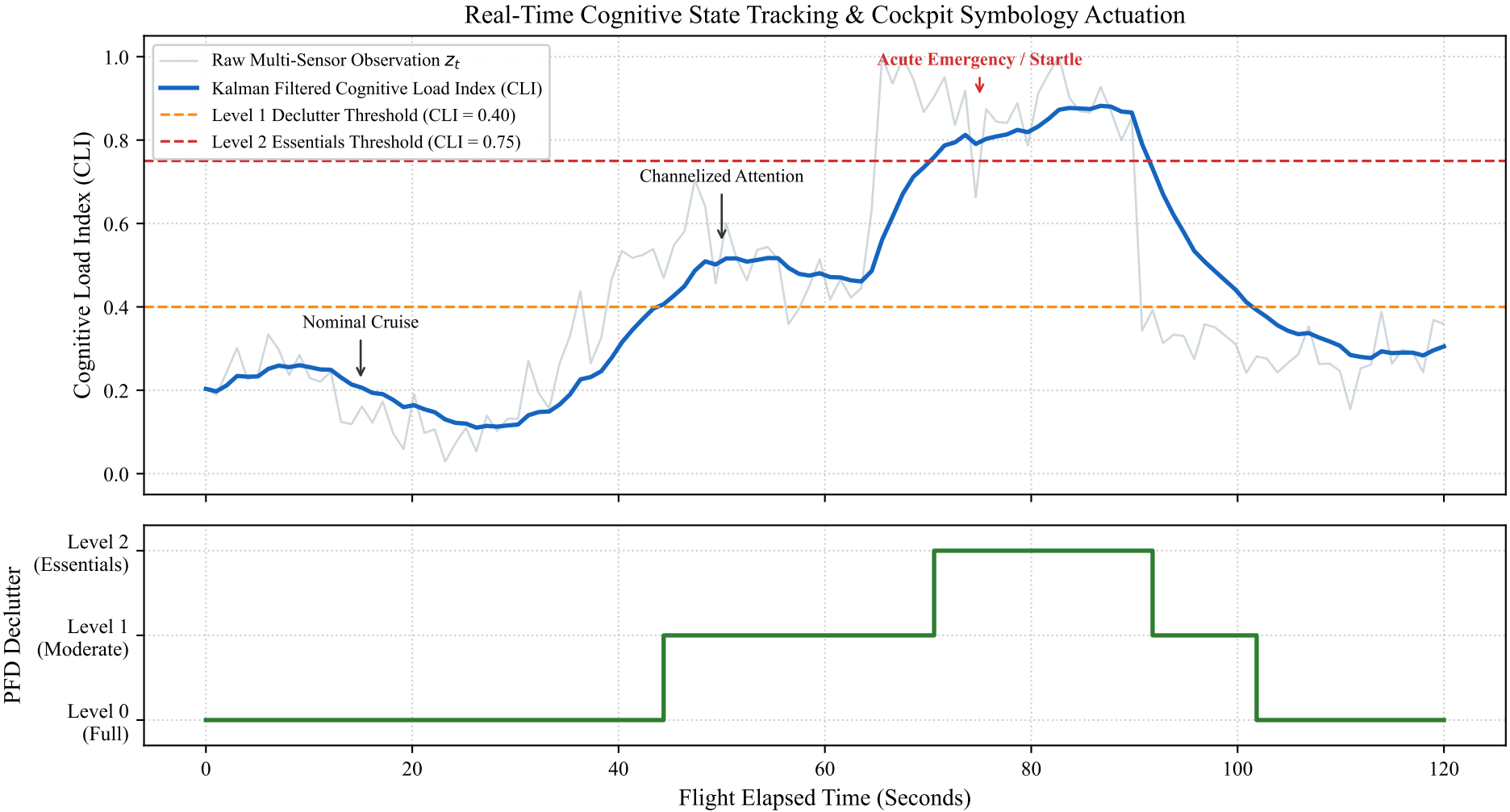
Real-Time Cognitive State Tracking & Cockpit Symbology Actuation
Raw Multi-Sensor Observation z_t
Kalman Filtered Cognitive Load Index (CLI)
Level 1 Declutter Threshold (CLI = 0.40)
Level 2 Essentials Threshold (CLI = 0.75)
Acute Emergency / Startle
Channelized Attention
Nominal Cruise
Cognitive Load Index (CLI)
1.0
0.8
0.6
0.4
0.2
0.0
PFD Declutter
Level 2 (Essentials)
Level 1 (Moderate)
Level 0 (Full)
0
20
40
60
80
100
120
Flight Elapsed Time (Seconds)